

Linear Algebra Overview Part 3

Ryan McLean

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1 Introduction

Now that we've seen what it is to be linear, and what a vector space is. Lets eluded to the idea of linear combinations and put some rigor behind it! Here we are going to discuss the mathematical representations used to work with these topics so that we can see how they work. This may make the ideas of linear algebra more confusing or maybe it will make them less confusing.

A couple of things, looking from this prospective I encourage you to use your imagination and try to build a vector space or linear system from the ideas discussed in this lecture notes. This is the foundation and I hope the foundation gives clarity to how the more abstract ideas of a vector space but it does take time and PRACTICE to see how these puzzle peaces fit together. I will provide examples but I highly encourage you if your struggling try to build a vector space! That is the best way to learn.

(Personally I took it to the extreme while learning and built from the ground up a vector space where the vectors and scalars were bananas)

2 Definitions and Theorems

Linear Combinations

Let V be a vector space with scalars in $\mathbb{R}$. Suppose $v_0, \dots, v_n \in V$ and $a_0, \dots, a_n \in \mathbb{R}$. Then a linear combinations is an expression of the form

$$a_0v_0 + a_1v_1 + \dots + a_nv_n$$

Observes, linear combinations are really just the addition of vectors with scalar weights.

Span

Let V be a vector space with scalars in $\mathbb{R}$. Suppose $v_0, \dots, v_n \in V$. Assume S is the set

$$S = \{v_0, v_1, \dots, v_n\}$$

Then the span of S is the set of all linear combinations, and is of the form

$$\text{span}(S) = \{a_0v_0 + a_1v_1 + \dots + a_nv_n \mid a_i \in \mathbb{R}, \forall 0 \leq i \leq n\}$$

Basis

Let V be a vector space with scalars in $\mathbb{R}$. A basis B for V is a subset $B \subset V$ such that the elements in B are linearly independent and span the vector space V .

Dimension

Let $B \subset V$ be a basis for the vector space V . The dimension of V is the number of vectors in B .

If B is the set $\{v_1, v_2, \dots, v_n\}$, then the dimension of V is n .

Any field works for the definition of a vector space, but rather than defining the technical details of what a field is I present the complex number which is a field, as is the real numbers. For more information on fields refer to Wikipedia

Linear Transform

Let V, W be vector spaces. Then $T : V \rightarrow W$ is a linear transform if the following hold for all $\vec{v}, \vec{u} \in V$ and $c \in \mathbb{R}$

$$T(c\vec{v}) = cT(\vec{v})$$

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

So what this is saying is that a linear transform is a function that respects the properties of linear combinations. That is to say the transform of a linear combination in V will remain a linear combination in W with the same weights. Or graphically, lines stay lines.

3 Examples

Things and stuff. Also important details. Lets discuss those important details by considering the example set of vectors S

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 16 \end{bmatrix}, \begin{bmatrix} 15 \\ 25 \\ 73 \end{bmatrix} \right\}$$

and its subsets.

First lets establish a naming convention for the subsets of S . I am going to label with subscripts the vectors in the subset of S listed in my subset as so:

$$S_{12} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix} \right\}$$

$$S_{124} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix}, \begin{bmatrix} 15 \\ 25 \\ 73 \end{bmatrix} \right\}$$

And so on. As you can see S_{12} has the first and second vector listed in S .

To discuss span lets look at the span of S_{123} , S_{124} , and S .

$$Span(S_{123}) = \{ \vec{v} \in \mathbb{R}^3 | \vec{v} = a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix} + a_3 \begin{bmatrix} 4 \\ 8 \\ 16 \end{bmatrix} \text{ where } a_1, a_2, a_3 \in \mathbb{R} \}$$

$$Span(S_{124}) = \{ \vec{v} \in \mathbb{R}^3 | \vec{v} = a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix} + a_4 \begin{bmatrix} 15 \\ 25 \\ 73 \end{bmatrix} \text{ where } a_1, a_2, a_4 \in \mathbb{R} \}$$

and

$$Span(S) = \{\vec{v} \in \mathbb{R}^3 | \vec{v} = a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 13 \\ 21 \\ 67 \end{bmatrix} + a_3 \begin{bmatrix} 4 \\ 8 \\ 16 \end{bmatrix} + a_4 \begin{bmatrix} 15 \\ 25 \\ 73 \end{bmatrix} \text{ where } a_1, a_2, a_3, a_4 \in \mathbb{R}\}$$

So what we are looking at is sets of linear combinations of vectors. Meaning we can plug in real numbers into our scalar weights to calculate elements in each set.

I am going to make two bold faced claims about these sets, and I leave it to you to justify them. I claim the subset S_{123} spans all of $\mathbb{R}^3$ and is thus equal to $\mathbb{R}^3$, and I will also claim the subset S_{124} dose not span all of $\mathbb{R}^3$ and is in fact a subset of $\mathbb{R}^3$.

To check these claims perform row operations on matrices that have the vectors as the columns. In these two examples that would be

$$\begin{bmatrix} 1 & 13 & 4 \\ 2 & 21 & 8 \\ 3 & 67 & 16 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 13 & 15 \\ 2 & 21 & 25 \\ 3 & 67 & 73 \end{bmatrix}$$

respectively for S_{123} and S_{124} .

If the result of applying row operations to put the matrices in reduced row echelon form is the identity matrix being

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

then the vectors span all of $\mathbb{R}^3$. If the resulting matrix is not the identity then they do not span all of $\mathbb{R}^3$. To convince you of this conclusion think of asking the question what scalar weights are needed to build a linear combination equal to any vector in $\mathbb{R}^3$? If you got the identity you have an answer to that question. If not that question has vectors that

have no solution to what those weights would be.

Now lets look at S . Well S has a subset S_{123} that spans $\mathbb{R}^3$ so S must span $\mathbb{R}^3$. But what about the forth vector? Well that has scalar weight zero in the span!

So lets look at basis. A basis must span the vector space and be linearly independent. Of our three examples that is only S_{123} as S_{124} is neither linearly independent nor does it span $\mathbb{R}^3$, just as S_{12} would not span all of $\mathbb{R}^3$. More S is not linearly independent so that disqualifies it from being a basis.

Building we can find the dimension of $\mathbb{R}^3$ to be 3 as there are 3 basis vectors in the basis we found.

Lets now look at a linear transform example.

The typical linear transform is a matrix applied to a coordinate vector. Thus lets consider the 2x2 matrix A

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the mapping $T(\vec{v}) = A\vec{v}$. Then we have

$$T(\vec{v}) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \vec{v}$$

Now to ensure this is a linear mapping we need to verify our requirements. Those being scalar multiplication and vector addition hold. So let $c \in \mathbb{R}$ and $\vec{u}, \vec{v} \in \mathbb{R}^2$. Looking first at scalar multiplication.

$$T(c\vec{v}) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} (c\vec{v})$$

But scalar multiplication is both associative and commutative. Meaning we can change where the parentheses are and we can move the c around the expression respectively. This means we can rewrite the equation as

$$T(c\vec{v}) = c \left(\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \vec{v} \right)$$

But that is

$$T(c\vec{v}) = cT(\vec{v})$$

so we have our requirement for scalar multiplication.

Lets check if our sums work.

$$T(\vec{v} + \vec{u}) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} (\vec{v} + \vec{u})$$

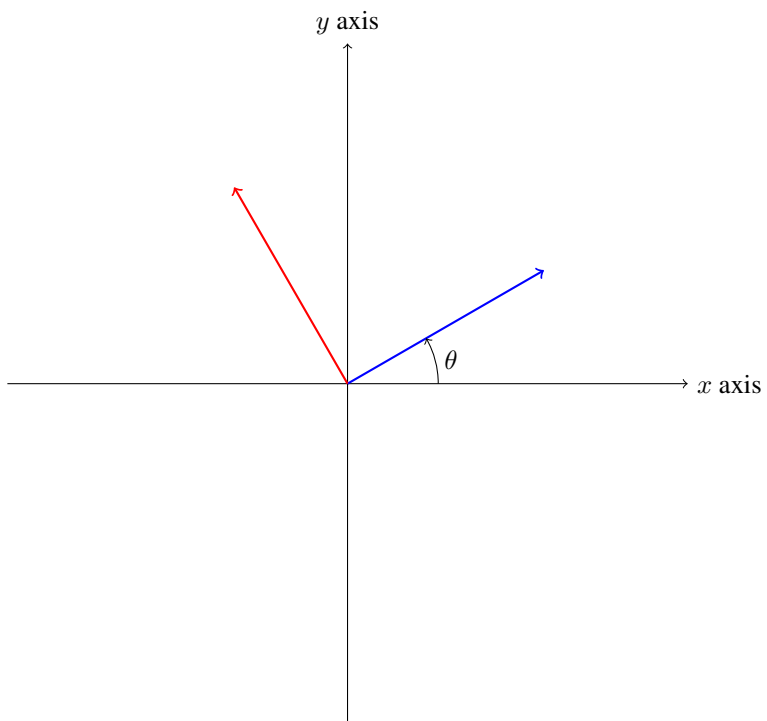
and as we know matrix multiplication distributes so we can write

$$T(\vec{v} + \vec{u}) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \vec{v} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \vec{u}$$

which is

$$T(\vec{v} + \vec{u}) = T(\vec{v}) + T(\vec{u})$$

giving us our requirement for addition, thus we know this transform is linear! But a bit more about what this transform is, this is the matrix for for rotating a 2 dimensional vector space by an angle θ .



The figure above shows where the basis vector $\hat{i}, \hat{j}$ go to under the transform T .